Roy Borkin

Industrial Designer | Programmer | Researcher | Maker

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Summary

With over a decade of experience in product development and cross-disciplinary research, I merge traditional craft with AI and computational design to solve complex challenges. Leveraging expertise across generative modeling, advanced manufacturing, and software development, I engineer data-driven physical products.

Highlights

13+ Experience

30+ Projects

8 Awards

3 Languages

Experience

Researcher

DesignTech Lab · Technion · Haifa | 2024-Current

Leads research in the fields of algorithms and AI, psychoacoustics, computational acoustics, generative design, marine biology and marine environments, artificial coral reef rehabitet, 3D printing, CNC, plastic recycling, and research-based design.

Lab Manager & Teaching Assistent

DesignTech Lab · Technion · Haifa | 2024-2026

The laboratory conducts research in design science, mechanics, AI, additive manufacturing, marine biology, medical science, and acoustics.
Oversee ongoing operations, manage budgets, coordinate collaborations and conferences, teaching Design and AI Courses, and ensure the maintenance of 3D printers and CNC machines.

Co-founder & Head of Product

SAFERA · California / Israel | 2023-2026

Co-founder of a startup specializing in weapon security solutions to prevent misuse and theft. Led development across mechanics, electronics, software, and design, delivering multiple functional POCs.

Mechanical Engineer & CAD for Manufacturing

Freelance

Have conceptualized and developed innovative products tailored for manufacturing. My expertise extends to creating detailed work plans, ensuring seamless production processes and optimal manufacturing.

Industrial & Mechanical designer

Jonathan Bar-Or studio · Pardes-Hanna | 2022-2024

Responsible for mechanical, medical equipment, and graphic designs.
As part of my role, I had to face customer’s issues and solve them using UX & UI, CAD and Procedural Modeling.

Workshop supervisor

Bezalel Academy · Jerusalem | 2019-2022

Trustee of the Industrial Design Workshops: wood, machining, plastics, and composite materials. CNC, laser, 3D print and manual machines operator.

Prosthesis Designer

Tikun Olam Makers · Tel Aviv | 2021

Product designer on a parametric prosthesis project.
In this project, I used my knowledge of programming and parametric design.

IDF - Industrial designer

Israel Defense Forces · 81 unit | 2013-2017

Designed, planned, and made functional products, provided services in industril design, professional material spraying, mechanicl engineering, UX & UI. Specialized in copying colors, surfaces, textures, materials & manufacturing technologies.

Education

PhD in Industrial Design

Technion · Haifa | 2026-Current

M.Sc in Industrial Design

Technion · Haifa | 2024-2026

Scuba Diver Researcher - AAUS

IUI · Eilat | 2026

Musical Instruments Craft

Living National Treasures · Tokyo | 2024

Fullstack Course

Code Chrysalis · Tokyo | 2024

B.Des in Industrial Design

Bezalel · Jerusalem | 2017-2022

CAM & CNC Operator

Modeling Center | 2020-2022

Safty Course for Workshops

Israel’s Department of Labor | 2019

CAD & Technical Drawing

IDF | 2015-2016

Skills

Software: Illustrator, Photoshop, AdobeXD, Figma, Premiere, Lightroom, InDesign, Unity, Unreal Engine, Office

3D Render & Model: SolidCAM, SolidWorks, Fusion360, Lumion, KeyShot, Blender, Rhino, GrassHopper, SketchUp, Zbrash, OnShape, TouchDesigner

Coding: C++, C#, JavaScript, HTML, CSS, SQL, Python, GCode, Local LLMs, Vibe Coding, n8n, Git, Node.js, Docker, ML, Deep Learning

Craft: Woodworking, Metalworking, Painting, Sculpturing, Luthiery (Guitars Building), Photography, 3D Printers Building, Scuba Diving

Personality: Motivated, Committed, Team-worker, Polite, Creative, Autodidact, Independent, Healpful, Active Listening, Attention to Detail

Languages: English, Hebrew, Japanese

Projects & Portfolio

https://drive.google.com/file/d/1OHgiktfbMpgeYK5bgWXIaBKXVmoINFFT/view?usp=sharing